Embedded_Plan

Subject	# Sessions	Outlines
ES Concepts	**2**	-Embedded Systems definition -Embedded Systems design -Embedded Systems Challenges -what is (IC , MPU , MCU , SOC , ECU) -Processor (CU , ALU , Reg Files) -Instruction Life Cycle -Instruction Set Architecture > RISC & CISC Processors -Register Files (GPR & SPR) == - Memory Types -Volatile (SRAM & DRAM) -Non-Volatile (Masked ROM & PROM & EPROM) -Hybrid (NVRAM & FLASH & EEPROM) - System Architecture -Von Neumann Architecture -Harvard Architecture -Input Output peripherals -Clock Systems (Electrical & Mechanical) □ Important Questions
MCU Fundamentals	**2**	-MCU Bus Interfaces (Master & Slave) -MCU Memmory Mapping -AMBA == -MCU Clocks -Reading the datasheet and specifications

MCU Interrupts	**2**	-Interrupt definition & Types -What is GIC/PIC ? -SW Mechanisms & Concepts -Servicing Interrupts & Vector Table Types -What is Startup Code ? -Interrupt Nesting & its types -Function Call vs SW INT -INT Concepts (Latency & Responce) -Instruction cycle with interrupts -Reading TRM for INT
GPIO & LED & SSD & SWs	**7**	-On&Off Chip peripheral -GPIO/DIO definition -GPIO Circuit & Block Diagram -Pin Modes -Input (Floating & Pullup & PullDown) -Output (PushPull & Open Drain) -AFIO -Led Interfacing -Sink & Source Connection -I/O Voltage Levels (DC C/CS) --> (TTL & CMOS) -ATMega32 Voltage Levels -AMR (Absolute Max Rating) -Introduction to layered architecture -OUR Naming Rule -DIO Driver for ATMega32 & STM32F103C6 -SSD Interfacing & POV -Electrical Switches -Transistors -Relays -Opto-couplers -Darlington Pair
Keypad	**1**	Keypad Interfacing and driver
LCD	**2**	-Display Types -LCD Internal Structure -LCD Interfacing and driver

AFIO & EXTI	**1**	-AFIO & its Registers -EXTI Driver

ADC	**4**	-Signals Intro
		-ADC definition
		-Digitalization Process
		-ADC Conceps & Types
		-ADC Types
		  -DAC Based Design (Ramp & SAR)
		  -Parallel Design (Flash)
		  -Integrator Based Design (Single & Dual Slope)
		  -Others (Sigma/Delta)
		-ADC Circuit (ATMeda32)
		-ADC Modes
		-ADC Registers & Driver
		-ADC Design Concepts (Synch & Asynch)
		-ADC Chain Conversion
		-Analog Sensors (LM35 &LDR)
		-Acuoustic Elements

TIMERS	**4**	-Timer definition & Concepts -Timer Modes -OV-F/Normal Mode (3 Cases) -CTC Mode -PWM (SW & HW) -ICU (SW & HW) -Watchdog timer □ Important Questions

Communication protocols intro	**1**	-Why Communication Protocols?
		-Communication Protocol Features & Categories
		  -Data Direction
		  -Transmission Teqniques
		  -Medium
		  -Synchronization
		  -N2N Relation
		  -Digital Levels
		  -Throuput
		-Bit/Data/Baud Rate
		-Wire/Line (Single Ended & Differential)

UART	**2**	-UART HW Connection -UART Specs -UART Frame -UART Way of Communication & SW Methods -UART Block Diagram -UART Registers & Driver
SPI	**2**	-SPI HW Connection -SPI Specs -SPI Frame -SPI Way of Communication & SW Methods -SPI Block Diagram -SPI Registers & Driver
TWI / I2C	**2**	-TWI HW Connection -TWI Specs -TWI Frame -TWI Way of Communication & SW Methods -TWI Block Diagram -TWI Registers & Driver -EEPROM Driver

RTOS	**4**	-Building Our Real time Scheduler -Real time operating systems concepts -Basic definitions -Scheduling Techniques -Dynamic Design Concepts -Shared Resources Analysis -Mutual exclusion Techniques -Inter task communications -Porting Free RTOS on AVR
Testing	**3**	- Embedded Systems Development Cycle - Basic Development Patterns & Models - Basic Testing Princeples - Verification vs Validation - Testing Types - Unit Testing - Module Testing - Integration Testing - Validation Testing - White & Black Box Testing & its Techniques Agile using Scrum Methodology
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